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MATH 111: Differential Calculus

Topic 9 Handout

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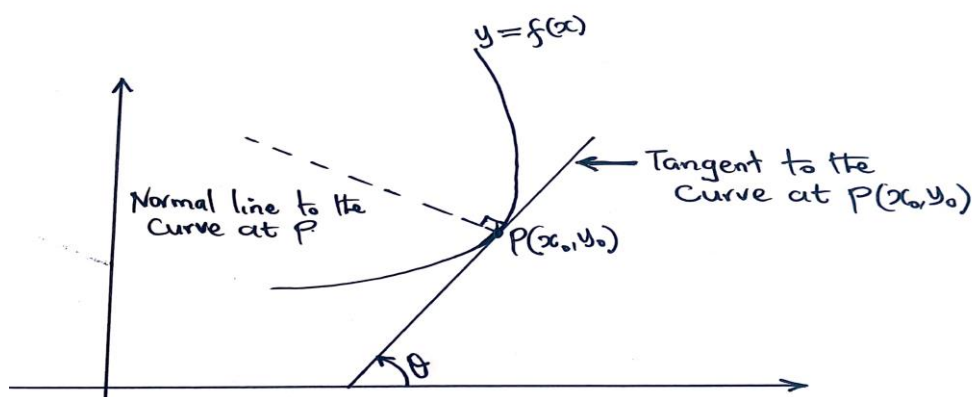
Published 2020

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Topic 9: Applications of Differentiation

9.1 Equations of Tangent and Normal lines to a curve

Consider the curve $y = f(x)$



The equation of a straight line passing through a given point $P(x_0, y_0)$ having finite slope m is given by $y - y_0 = m(x - x_0)$

The slope of the tangent line to the curve $y = f(x)$ at the point $P(x_0, y_0)$ is given by $f'(x_0) = m$

The equation of the tangent line at $P(x_0, y_0)$ to the curve $y = f(x)$ is given by

$$y - y_0 = m(x - x_0)$$

Since the normal line is perpendicular to the tangent line, the slope of the normal line to the curve

$y = f(x)$ at the point $P(x_0, y_0)$ is $-\frac{1}{m}, m \neq 0$

The equation of the normal line to the curve $y = f(x)$ at the point $P(x_0, y_0)$ is given by

$$y - y_0 = -\frac{1}{m}(x - x_0)$$

$$\Rightarrow m(y - y_0) + (x - x_0) = 0$$

Note: If a tangent line to the curve $y = f(x)$ makes an angle θ with the x -axis in the positive direction, then $m = \tan \theta$.

Example 1

Find the point at which the tangent line to the curve $y = \sqrt{4x-3} - 1$ has its slope as $\frac{2}{3}$.

Solution

$$y = (4x-3)^{\frac{1}{2}} - 1$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}(4x-3)^{-\frac{1}{2}} \cdot 4 \\ &= \frac{2}{\sqrt{4x-3}}\end{aligned}$$

But the slope is given as $\frac{2}{3}$

$$\Rightarrow \frac{2}{\sqrt{4x-3}} = \frac{2}{3}$$

$$\Rightarrow 4x-3=9$$

$$\Rightarrow x=3$$

When $x=3$,

$$\begin{aligned}y &= \sqrt{12-3} - 1 \\ &= 2\end{aligned}$$

Therefore, the point is $(3,2)$.

Example 2

Find the equations of tangent line and normal line to the curve defined by $x = 2t - t^2$, $y = 3t - t^3$ which passes through the origin.

Solution

Solving the equations at the origin where $x = 0$, $y = 0$

$$t(2-t) = 0$$

$$\Rightarrow t = 0, \quad t = 2$$

$$t(3-t^2) = 0$$

$$\Rightarrow t = 0, \quad t = \pm\sqrt{3}$$

Therefore, at the origin, $t = 0$

$$\frac{dx}{dt} = 2 - 2t$$

$$\frac{dy}{dt} = 3 - 3t^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3-3t^2}{2-2t}$$

At $t = 0$, $\frac{dy}{dx} = \frac{3}{2}$

Equation of the tangent line is given by

$$y - 0 = \frac{3}{2}(x - 0)$$

$$\Rightarrow y = \frac{3}{2}x$$

Equation of the normal line is given by

$$y - 0 = -\frac{2}{3}(x - 0)$$

$$\Rightarrow y = -\frac{2}{3}x$$

Example 3

Find the equations of the tangent line and normal line to the curve $y = \tan^{-1}\left(\frac{x}{2}\right)$ at $x = 3$.

Solution

$$y = \tan^{-1}\left(\frac{x}{2}\right)$$

$$\frac{dy}{dx} = \frac{1}{1 + \left(\frac{x}{2}\right)^2} \cdot \frac{1}{2}$$

$$= \frac{0.5}{1 + \left(\frac{x}{2}\right)^2}$$

At $x = 3$,

$$\frac{dy}{dx} = \frac{0.5}{1 + (1.5)^2}$$

$$= 0.1538$$

But when $x = 3$, $y = \tan^{-1}(1.5) = 0.9828$

At the point $(3, 0.9828)$, $m = 0.1538$

Therefore, the equation of the tangent line to the curve is given by

$$y - 0.9828 = 0.1538(x - 3)$$

$$\Rightarrow y = 0.1538x + 0.5214$$

The equation of the normal line to the curve is given by

$$y - 0.9828 = -\frac{1}{0.1538}(x - 3)$$

$$= -6.5x + 19.5$$

$$\Rightarrow y = -6.5x + 20.48$$

9.2 Kinematics

Displacement: This is the position function of an object, usually denoted by $S(t)$. It is the directed distance of an object from the origin at a time t .

Velocity: This is the rate of change of displacement with respect to time. The velocity function $V(t)$ is the derivative of the position function i.e.

$$V = S'(t) = \frac{dS}{dt}$$

Acceleration: This is the rate of change of velocity with respect to time. The acceleration $a(t)$ is the derivative of the velocity function i.e.

$$a = V'(t) = \frac{dV}{dt} = \frac{d^2S}{dt^2}$$

Remark:

a) Positive acceleration refers to increasing velocity i.e. $a = \frac{dv}{dt} > 0$

b) Negative acceleration refers to decreasing velocity i.e. $a = \frac{dv}{dt} < 0$

c) At zero velocity, ($V=0$), the object is said to be stationary and at this point, a maximum height is achieved.

- d) At zero acceleration ($a=0$), the object is said to have a constant velocity and at this point, maximum velocity is achieved.

Note:

- i. If $a(t) \cdot v(t) > 0$, then the object is speeding up at a time t .
- ii. If $a(t) \cdot v(t) < 0$, then the object is slowing down at a time t .
- iii. If $s(t) \cdot v(t) > 0$, then object is moving away from the origin at a time t .
- iv. If $s(t) \cdot v(t) < 0$, then object is moving towards the origin at a time t .

Example 1

A particle is moving on a straight line and its distance S meters from a fixed point in the line after t seconds is given by $S = 12t - 15t^2 + 4t^3$. Find:

- a) The velocity and acceleration of the particle after 3 sec.

Solution

$$S = 12t - 15t^2 + 4t^3$$

$$V = \frac{ds}{dt}$$

$$= 12 - 30t + 12t^2$$

At $t = 3$,

$$V = 12 - 30(3) + 12(3)^2$$

$$= 30 \text{ m/s}$$

$$a = \frac{dV}{dt}$$

$$= -30 + 24t$$

At $t = 3\text{sec}$,

$$a = -30 + 24(3)$$

$$= 42\text{m/s}^2$$

- b) The distance travelled between the two points when the velocity is instantaneously zero.

Solution

When velocity is zero,

$$\Rightarrow 12 - 30t + 12t^2 = 0$$

$$\Rightarrow (2t - 1)(t - 2) = 0$$

Velocity is zero at the points when $t = \frac{1}{2}$, $t = 2$

When $t = \frac{1}{2}$,

$$S = 12\left(\frac{1}{2}\right) - 15\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right)^3$$

$$= 2.75\text{m}$$

When $t = 2$,

$$S = 12(2) - 15(2)^2 + 4(2)^3$$

$$= -4\text{m}$$

Hence the distance travelled between $t = \frac{1}{2}$ and $t = 2$ is given as

$$\begin{aligned} S &= |2.75| + |-4| \\ &= 6.75m \end{aligned}$$

Example 2

A ball thrown vertically upwards attains a height of $S = 30t - 5t^2$ after t seconds. Calculate:

- a) The initial velocity of the ball.

Solution

$$\begin{aligned} V &= \frac{ds}{dt} \\ &= 30 - 10t \end{aligned}$$

Initial velocity is the velocity at $t = 0$

At $t = 0$,

$$\begin{aligned} V &= 30 - 10(0) \\ &= 30m/s \end{aligned}$$

- b) Velocity of the ball after 1sec.

Solution

$$V = 30 - 10t$$

At $t = 1$,

$$V = 30 - 10(1)$$

$$= 20 \text{ m/s}$$

c) Maximum height reached by the ball.

Solution

At maximum height, $V = 0 \text{ m/s}$

$$\Rightarrow 30 - 10t = 0$$

$$\Rightarrow t = 3 \text{ sec}$$

Therefore at $t = 3 \text{ sec}$, the ball reaches a maximum height

$$\begin{aligned} S_{\max} &= 30t - 5t^2 \\ &= 30(3) - 5(3)^2 \\ &= 45 \text{ m} \end{aligned}$$

Example 3

The displacement of an object at any time $t \text{ sec}$ is given by $S = 6t^2 - 2t^3 \text{ m}$.

a) Find the time when the object comes to rest.

Solution

At rest, $V = 0 \text{ m/s}$

$$S = 6t^2 - 2t^3$$

$$V = \frac{ds}{dt}$$

$$= 12t - 6t^2$$

When $V = 0$,

$$\Rightarrow 12t - 6t^2 = 0$$

$$t(12 - 6t) = 0$$

$$\Rightarrow t = 0, \quad t = 2$$

$$\therefore \underline{t = 2 \text{ sec.}}$$

b) Show that the object is moving away from the origin at $t = 1$.

Solution

$$S(1) = 6(1)^2 - 2(1)^3$$

$$= 4$$

$$V(1) = 12(1) - 6(1)^2$$

$$= 6$$

$$S(1)V(1) = 6(4)$$

$$= 24 > 0$$

Since $S(1)V(1) > 0$, then the object is moving away from the origin.

9.3 Related Rates of Change

Considering two variables x and y which are both changing with time. A “related rates” problem is a problem in which one of the rates of change at a given instant is known and the other rate at that

instant is to be determined. We find the equation that relates the two quantities and then use the appropriate technique of differentiation.

Example 1

A circular disc of radius 3cm is being heated. Due to expansion, its radius increases at the rate of 0.05cm/s . Find the rate at which its area is increasing when the radius is 3.2cm .

Solution

Area, $A = \pi r^2$

Given that $\frac{dr}{dt} = 0.05$

We then have $\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$

But $\frac{dA}{dr} = 2\pi r$

Therefore

$$\begin{aligned}\frac{dA}{dt} &= 2\pi r(0.05) \\ &= 0.1\pi r\end{aligned}$$

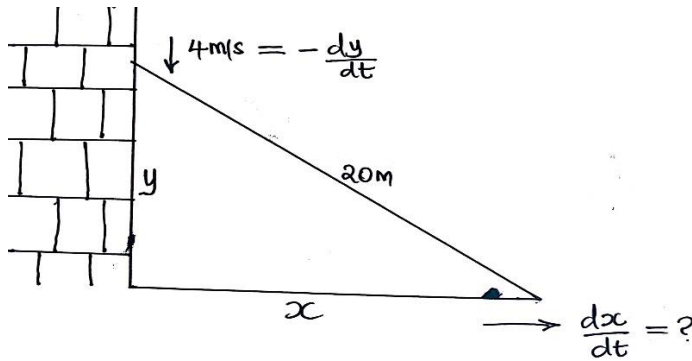
When $r = 3.2$,

$$\begin{aligned}\frac{dA}{dt} &= 0.1\pi(3.2) \\ &= 0.32\pi\text{cm}^2/\text{s}\end{aligned}$$

Example 2

A $20m$ ladder leans against a wall. The top of the ladder slides down at the rate of $4m/s$. How fast is the bottom of the ladder moving when it is $16m$ from the wall?

Solution



The relation between x and y is $x^2 + y^2 = 20^2$

$$\frac{d}{dt}(x^2) + \frac{d}{dt}(y^2) = \frac{d}{dt}(400)$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

But $\frac{dy}{dt} = -4m/s$

$$\Rightarrow \frac{dx}{dt} = \frac{-2y(-4)}{2x}$$

$$= \frac{4y}{x}$$

When $x = 16$,

$$y = \sqrt{20^2 - 16^2}$$

$$= 12$$

Therefore,

$$\frac{dx}{dt} = \frac{4(12)}{16}$$

$$= 3m / s$$

Example 3

The length $x\text{cm}$ of a rectangle is decreasing at the rate of $3\text{cm}/\text{min}$ and its width $y\text{cm}$ is increasing at the rate of $2\text{cm}/\text{min}$. When $x=10\text{cm}$ and $y=6\text{cm}$, find the rate of change of:

- a) Perimeter of the rectangle.

Solution

$$\frac{dx}{dt} = -3\text{cm}/\text{min},$$

$$\frac{dy}{dt} = 2\text{cm}/\text{min}$$

Perimeter,

$$P = 2(x + y)$$

$$\begin{aligned}\frac{dP}{dt} &= 2\left(\frac{dx}{dt} + \frac{dy}{dt}\right) \\ &= 2(-3 + 2) \\ &= -2\text{cm}/\text{min}\end{aligned}$$

Therefore, the perimeter decreases at the rate of $2\text{cm}/\text{min}$

- b) Area of the rectangle.

Solution

$$\text{Area, } A = x \cdot y$$

$$\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$$

$$= 10(2) + 6(-3)$$

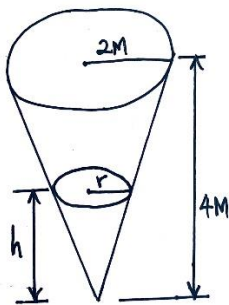
$$= 2 \text{ cm}^2 / \text{min}$$

Therefore, the area increases at the rate of $2 \text{ cm}^2 / \text{min}$.

Example 4

A water tank has the shape of an inverted circular cone with base radius 2m and height 4m . If the water is being pumped into the tank at the rate of $2 \text{ m}^3 / \text{min}$. Find the rate at which the water is rising when the water is 3m deep.

Solution



We find $\frac{dh}{dt}$ given that $\frac{dV}{dt} = 2$

By similarity,

$$\frac{h}{r} = \frac{4}{2}$$

$$\Rightarrow r = \frac{h}{2}$$

But

$$V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h$$

$$= \frac{1}{12}\pi h^3$$

$$\frac{dV}{dh} = \frac{1}{4}\pi h^2$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$2 = \frac{\pi h^2}{4} \cdot \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{2}{\frac{\pi h^2}{4}}$$

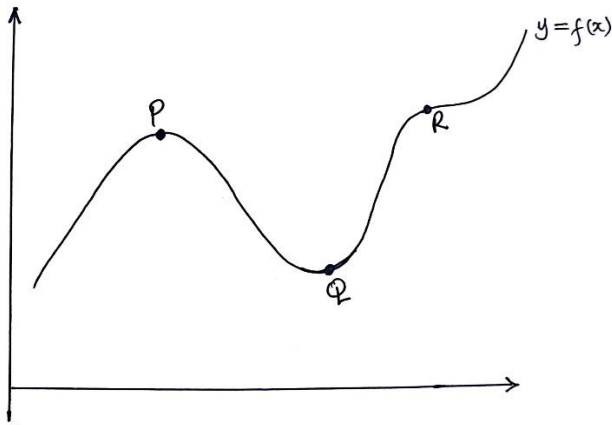
$$= \frac{8}{\pi h^2}$$

But $h = 3$,

$$\Rightarrow \frac{dh}{dt} = \frac{8}{9\pi} \text{ m/min}$$

9.4 Maximum and Minimum Values

Consider the curve $y = f(x)$ with the following concaves.



Concave downwards (P), the turning point is called **a maximum point**. At every maximum point, $\frac{dy}{dx} = 0$.

On the left-hand side of a maximum point, $\frac{dy}{dx} > 0$ i.e. positive

On the right-hand side of a maximum point, $\frac{dy}{dx} < 0$ i.e. negative

x			P		
$\frac{dy}{dx}$	+	ve	0	-	ve
	↗		—		↘

Concave upwards (Q), the turning point is called **a minimum point**. At every minimum point, $\frac{dy}{dx} = 0$

On the left-hand side of a minimum point, $\frac{dy}{dx} < 0$ i.e. negative

On the right-hand side of a minimum point, $\frac{dy}{dx} > 0$ i.e. positive

x		Q	
$\frac{dy}{dx}$	$-ve$	0	$+ve$
			

At a point R , the curve has a turning point called **a point of inflexion**. Therefore R is a point of inflexion if the sign before and after the turning point does not change.

x		R	
$\frac{dy}{dx}$	$+ve$	0	$+ve$
			

OR

x		R	
$\frac{dy}{dx}$	$-ve$	0	$-ve$
			

Note: A point x_0 is said to be a critical point if the derivative $f'(x_0) = 0$. Maximum points, Minimum points and points of inflexions are called **extremum**

Example 1

Find and classify all the turning points of the curve $y = x^3 + 3x^2 - 9x + 2$.

Solution

$$y = x^3 + 3x^2 - 9x + 2$$

$$\frac{dy}{dx} = 3x^2 + 6x - 9$$

At every turning point, $\frac{dy}{dx} = 0$

$$\Rightarrow 3x^2 + 6x - 9 = 0$$

$$(x-1)(x+3) = 0$$

$$\Rightarrow x = 1, \quad x = -3$$

When $x = 1$,

$$y = 1^3 + 3(1)^2 - 9(1) + 2$$

$$= -3$$

When $x = -3$,

$$y = (-3)^3 + 3(-3)^2 - 9(-3) + 2$$

$$= 29$$

The turning points are $(1, -3)$ and $(-3, 29)$

Classifying the turning points;

For the point $(1, -3)$, we have

x	0	1	2
$\frac{dy}{dx}$	-ve	0	+ve
			

$\Rightarrow (1, -3)$ is a minimum point.

For the point $(-3, 29)$, we have

x	-4	-3	0
$\frac{dy}{dx}$	+	0	-
	(	-	)

$\Rightarrow (-3, 29)$ is a maximum point

Second Derivative Test for Turning Points

We define $\frac{d^2y}{dx^2}$ as the derivative of $\frac{dy}{dx}$ i.e. $\frac{d^2y}{dx^2} = \frac{d}{dx} \left[\frac{dy}{dx} \right]$

e.g. If $y = x^2 - \frac{1}{x}$, then

$$\frac{dy}{dx} = 2x + \frac{1}{x^2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} (2x + x^{-2})$$

$$= 2 - \frac{2}{x^3}$$

Suppose $(x_0, f(x_0))$ is a turning point of $f(x)$ and if:

a) $\frac{d^2y}{dx^2} < 0$, then $(x_0, f(x_0))$ is a maximum turning point.

b) $\frac{d^2y}{dx^2} > 0$, then $(x_0, f(x_0))$ is a minimum turning point.

c) $\frac{d^2y}{dx^2} = 0$, then the type of the turning point is determined by the first derivative (by finding the sign of the gradient on each side of the point).

Note: Points of inflection are hidden turning points which occurs when $\frac{d^2y}{dx^2} = 0$.

Example1

Find and classify the turning points of the following curves using the second derivative test:

a) $y = \frac{1}{3}x^3 - 2x^2 + 3x - 2$

Solution

$$\frac{dy}{dx} = x^2 - 4x + 3$$

At every turning point, $\frac{dy}{dx} = 0$

$$\Rightarrow x^2 - 4x + 3 = 0$$

$$\Rightarrow (x-1)(x-3) = 0$$

$$\Rightarrow x = 1, \quad x = 3$$

When $x = 1$,

$$y = \frac{1}{3}(1)^3 - 2(1)^2 + 3(1) - 2$$

$$= -\frac{2}{3}$$

When $x = 3$,

$$y = \frac{1}{3}(3)^3 - 2(3)^2 + 3(3) - 2$$

$$= -2$$

Therefore, the turning points are: $\left(1, -\frac{2}{3}\right)$ and $(3, -2)$

Using the second derivative to classify the turning points,

$$\frac{d^2y}{dx^2} = 2x - 4$$

At $x=1$,

$$\frac{d^2y}{dx^2} = 2(1) - 4$$

$$= -2 < 0$$

$\Rightarrow \left(1, -\frac{2}{3}\right)$ is a maximum point

At $x=3$,

$$\frac{d^2y}{dx^2} = 2(3) - 4$$

$$= 2 > 0$$

$\Rightarrow (3, -2)$ is a minimum point

b) $y = x^3 + 9x^2 + 15x - 4$

Solution

$$\frac{dy}{dx} = 3x^2 + 18x + 15$$

At every turning point, $\frac{dy}{dx} = 0$

$$\Rightarrow 3x^2 + 18x + 15 = 0$$

$$(x+1)(x+5) = 0$$

$$\Rightarrow x = -1, \quad x = -5$$

When $x = -1$,

$$\begin{aligned} y &= (-1)^3 + 9(-1)^2 + 15(-1) - 4 \\ &= -11 \end{aligned}$$

When $x = -5$,

$$\begin{aligned} y &= (-5)^3 + 9(-5)^2 + 15(-5) - 4 \\ &= 21 \end{aligned}$$

Turning points are: $(-1, -11)$ and $(-5, 21)$

Using the second derivative to classify the turning points

$$\frac{d^2y}{dx^2} = 6x + 18$$

At $(-1, -11)$,

$$\begin{aligned} \frac{d^2y}{dx^2} &= -6 + 18 \\ &= 12 > 0 \end{aligned}$$

$\Rightarrow (-1, -11)$ is a minimum point.

At $(-5, 21)$,

$$\begin{aligned}\frac{d^2y}{dx^2} &= -30 + 18 \\ &= -12 < 0\end{aligned}$$

$\Rightarrow (-5, 21)$ is a maximum point

Example 2

Find all the point(s) of inflection for the given curve $y = (x^2 - 3)^2$.

Solution

$$y = (x^2 - 3)^2$$

$$\begin{aligned}\frac{dy}{dx} &= 2(x^2 - 3) \cdot 2x \\ &= 4(x^3 - 3x)\end{aligned}$$

$$\frac{d^2y}{dx^2} = 4(3x^2 - 3)$$

But points of inflection occur when $\frac{d^2y}{dx^2} = 0$

$$\begin{aligned}\Rightarrow 12(x^2 - 1) &= 0 \\ (x - 1)(x + 1) &= 0 \\ \Rightarrow x &= -1, \quad x = 1\end{aligned}$$

When $x = -1$,

$$\begin{aligned}y &= ((-1)^2 - 3)^2 \\ &= 4\end{aligned}$$

When $x = 1$,

$$y = ((1)^2 - 3)^2 \\ = 4$$

The points of inflections are $(-1, 4)$ and $(1, 4)$.

9.5 Curve Sketching

In curve sketching, we find the general course of the curve by marking;

- (i) All the turning points (where $\frac{dy}{dx} = 0$).
- (ii) Intercepts ($x=0$ gives y -intercept and $y=0$ gives x -intercept).

Example 1

Sketch the curve $y = x^3 - 3x^2 - 9x$.

Solution

- (i) Turning points

$$y = x^3 - 3x^2 - 9x$$

$$\frac{dy}{dx} = 3x^2 - 6x - 9$$

At every turning point, $\frac{dy}{dx} = 0$

$$\Rightarrow 3x^2 - 6x - 9 = 0$$

$$(x+1)(x-3) = 0$$

$$\Rightarrow x = -1, \quad x = 3$$

When $x = -1$,

$$\begin{aligned}y &= (-1)^3 - 3(-1)^2 - 9(-1) \\&= 5\end{aligned}$$

When $x = 3$,

$$\begin{aligned}y &= (3)^3 - 3(3)^2 - 9(3) \\&= -27\end{aligned}$$

Turning points are; $(-1, 5)$ and $(3, -27)$

Classifying the turning points;

$$\frac{d^2y}{dx^2} = 6x - 6$$

At $x = -1$,

$$\begin{aligned}\frac{d^2y}{dx^2} &= 6(-1) - 6 \\&= -12 < 0\end{aligned}$$

$\Rightarrow (-1, 5)$ is a maximum point

At $x = 3$,

$$\begin{aligned}\frac{d^2y}{dx^2} &= 6(3) - 6 \\&= 12 > 0\end{aligned}$$

$\Rightarrow (3, -27)$ is a minimum point

(ii) Intercepts

When $x = 0$, $y = 0$

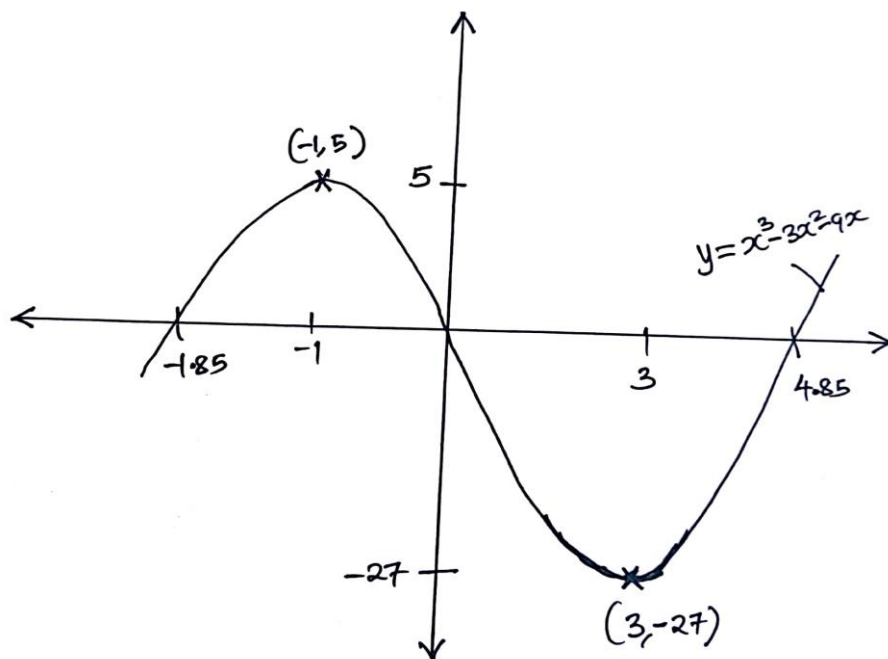
When $y = 0$,

$$\Rightarrow x(x^2 - 3x - 9) = 0$$

$$\Rightarrow x = 0 \text{ or } x^2 - 3x - 9 = 0$$

$$\Rightarrow x = 0, \quad x = -1.85, \quad x = 4.85$$

Therefore, we sketch the curve as follows.



Example 2

Sketch the curve $y = 3x - x^3$.

Solution

$$y = 3x - x^3$$

$$\frac{dy}{dx} = 3 - 3x^2$$

At every turning point, $\frac{dy}{dx} = 0$

$$\Rightarrow 3 - 3x^2 = 0$$

$$\Rightarrow (1-x)(1+x) = 0$$

$$\Rightarrow x = -1, \quad x = 1$$

When $x = -1$,

$$\begin{aligned} y &= 3(-1) - (-1)^3 \\ &= -2 \end{aligned}$$

When $x = 1$,

$$\begin{aligned} y &= 3(1) - (1)^3 \\ &= 2 \end{aligned}$$

Turning points are; $(-1, -2)$ and $(1, 2)$

Classifying the turning points;

$$\frac{d^2y}{dx^2} = -6x$$

At $x = -1$,

$$\begin{aligned} \frac{d^2y}{dx^2} &= -6(-1) \\ &= 6 > 0 \end{aligned}$$

$\Rightarrow (-1, -2)$ is a minimum point

At $x = 1$,

$$\begin{aligned}\frac{d^2y}{dx^2} &= -6(1) \\ &= -6 < 0\end{aligned}$$

$\Rightarrow (1, 2)$ is a maximum point

Intercepts

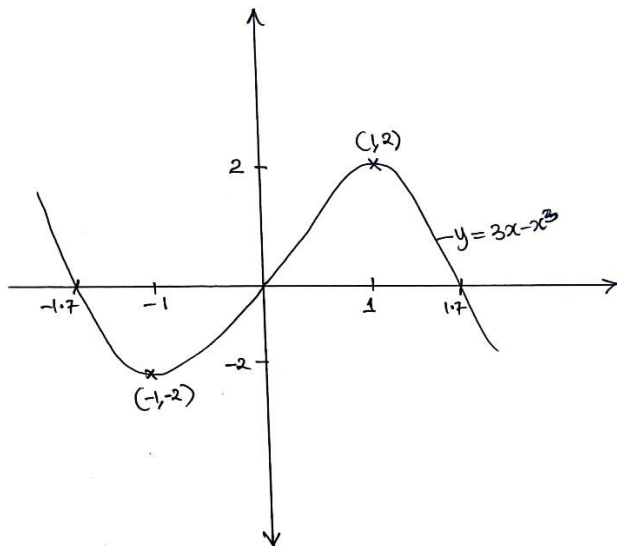
When $x = 0$, $y = 0$

When $y = 0$, $3x^2 - x^3 = 0$

$$\Rightarrow x(3 - x^2) = 0$$

$$\Rightarrow x = 0, \quad x = -1.7, \quad x = 1.7$$

Using the obtained information, we sketch the curve as follows.



Note: To find intercepts for functions such as $y = x^3 + x + 2$, we determine the factors of the constant term that will generate zero upon substitution on the function y , then synthesize using long method of division as shown.

For $x = -1$, $y = 0$ hence $(x+1)$ is a factor thus $\frac{x^3 + x + 2}{x+1} = x^2 - x + 2$

Thus $x^3 + x + 2$ can be factored into $(x+1)(x^2 - x + 2)$

$$\Rightarrow x = -1, \quad x^2 - x + 2 = 0$$

The quadratic equation can then be solved using any convenient method.

Asymptotes

An asymptote is an imaginary barrier that the function can approach but never touches or crosses it. It gives the behavior of the curve at infinity.

Types of asymptotes

Vertical asymptote (V.A)

This is obtained by setting the denominator equal to zero i.e If $\lim_{x \rightarrow a} f(x) = \pm\infty$, then $y = a$ is called a vertical asymptote.

To obtain the vertical asymptote:

- (i) Factor the denominator and numerator if possible.
- (ii) Cancel the common factor.
- (iii) Denominator that cancel completely give rise to “holes”. Those that don’t give rise to vertical asymptote.

Example 1

$$y = \frac{3x^3 - 8x^2 - 87x - 28}{x^3 - 10x^2 - 7x + 196}$$

$$\Rightarrow y = \frac{(3x+1)(x-7)(x+4)}{(x-7)^2(x+4)}$$

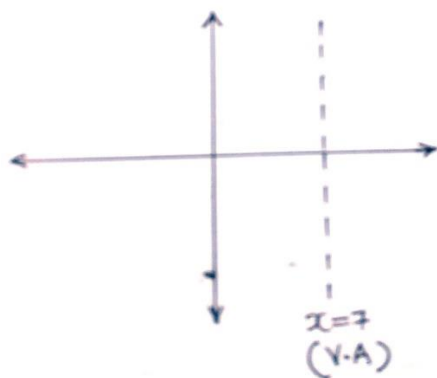
Cancelling out the common factors, we have

$$y = \frac{3x+1}{x-7}, \quad x \neq -4$$

Therefore $(x+4)$ cancel completely $\Rightarrow$ a hole at $x = -4$

Vertical asymptote, V.A;

$$\begin{aligned} x-7 &= 0 \\ x &= 7 \end{aligned}$$



Example 2

$$f(x) = \frac{1}{x-4}$$

The vertical asymptote, V.A is given by

$$\begin{aligned} x-4 &= 0 \\ x &= 4 \end{aligned}$$

Horizontal asymptote (H.A)

Look at the degree of the numerator and denominator, then:

1. If the degree at the top < degree at the bottom, then $y = 0$ is the horizontal asymptote

$$\text{i.e. } y = \frac{x^2}{x^3 - 1}$$

$$\Rightarrow y = 0; \text{ H.A}$$

2. If the degree at the top equals to the degree at the bottom, then the quotient of the coefficient of the highest degree terms make up the horizontal asymptote.

$$\text{i.e. } y = \frac{4x^2 + 11}{1 - 2x^2}$$

$$\Rightarrow y = \frac{4}{-2} = -2; \text{ H.A}$$

Note: If $\lim_{x \rightarrow \infty} f(x) = k$, then $y = k$ is a horizontal asymptote

3. If the top degree > degree at the bottom by one factor, then we obtain **oblique or slanting asymptote**.

To obtain slant asymptote, we use long method of division, whatever is obtain, throw away the remainder term. What is left is the slant asymptote.

$$\text{i.e. } y = \frac{x^2 - 1}{x}$$

Therefore

$$y = \frac{x^2 - 1}{x}$$

$$= x - \frac{1}{x}$$

Hence $y = x$ is the slant asymptote

Example 1

Sketch the following curve $y = \frac{x^2}{x - 1}$

Solution

Turning points

$$\begin{aligned}\frac{dy}{dx} &= \frac{2x(x-1) - x^2(1)}{(x-1)^2} \\ &= \frac{x^2 - 2x}{(x-1)^2}\end{aligned}$$

At every turning point, $\frac{dy}{dx} = 0$

$$\begin{aligned}\Rightarrow \frac{x^2 - 2x}{(x-1)^2} &= 0 \\ \Rightarrow x^2 - 2x &= 0 \\ \Rightarrow x = 0, x = 2\end{aligned}$$

When $x = 0$, $y = 0$

When $x = 2$, $y = 4$

Turning points are $(0,0)$ and $(2,4)$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{(2x-2)(x-1)^2 - (x^2-2x) \cdot 2(x-1) \cdot 1}{(x-1)^4} \\ &= \frac{2}{(x-1)^3}\end{aligned}$$

At $(0,0)$,

$$\frac{d^2y}{dx^2} = -2 < 0$$

$\Rightarrow (0,0)$ is a maximum point

At $(2,4)$,

$$\frac{d^2y}{dx^2} = 2 > 0$$

$\Rightarrow (2,4)$ is a minimum point

Intercepts

When $x=0$, $y=0$

Asymptotes

$$y = \frac{x^2}{x-1}$$

$$\text{V.A; } x-1=0$$

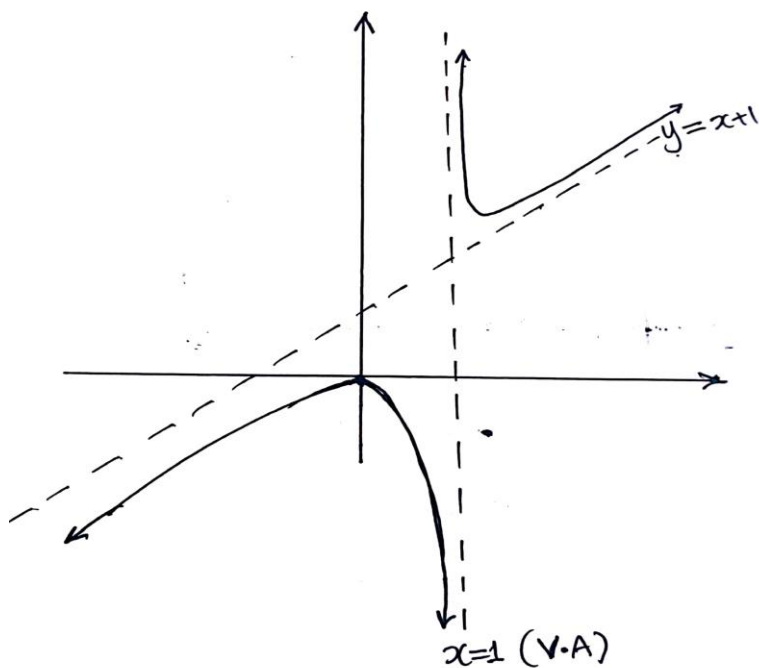
$$\Rightarrow x=1$$

S.A;

$$\begin{aligned} y &= \frac{x^2}{x-1} \\ &= x+1 + \frac{1}{x-1} \end{aligned}$$

$$\therefore y = x+1; \text{ S.A}$$

Sketching the curve, we obtain



Example 2

Sketch the curve $y = \frac{5-3x^2}{1-x^2}$.

Solution

Turning points

$$\frac{dy}{dx} = \frac{(1-x^2) \cdot (-6x) - (5-3x^2) \cdot (-2x)}{(1-x^2)^2}$$

$$= \frac{4x}{(1-x^2)^2}$$

At every turning point, $\frac{dy}{dx} = 0$

$$\Rightarrow \frac{4x}{(1-x^2)^2} = 0$$

$$\Rightarrow x=0$$

When $x=0$, $y=5$

Turning point is $(0,5)$

Classifying the turning point

x	-5	0	10
$\frac{dy}{dx}$	-ve	0	+ve
			

$\therefore (0,5)$ is a minimum point

Intercepts

When $x=0$, $y=5$

When $y=0$, $\Rightarrow 5-3x^2=0$

$$\Rightarrow 3x^2=5$$

$$\Rightarrow x=\pm 1.3$$

Asymptotes

V.A;

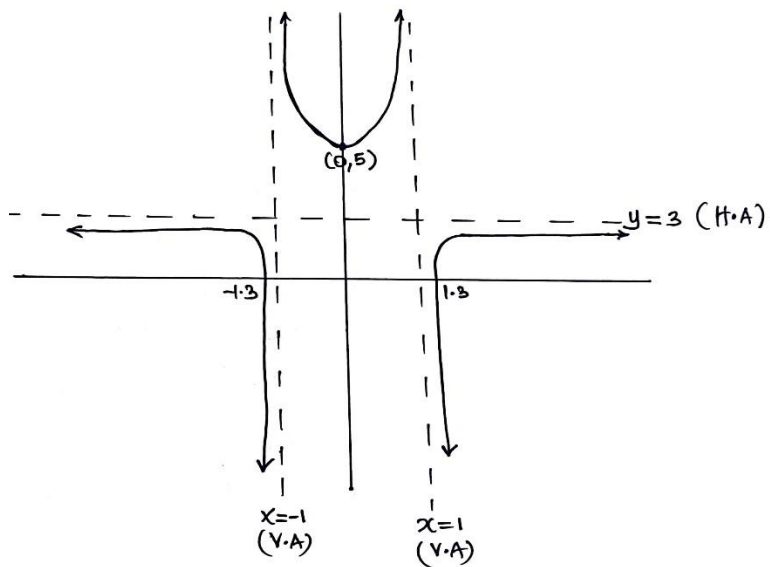
$$1-x^2=0$$

$$\Rightarrow x=\pm 1$$

H.A;

$$y = \frac{-3}{-1}$$

$$= 3$$



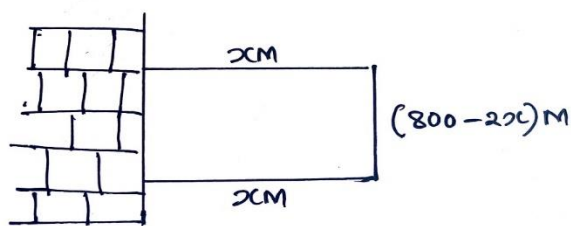
9.6 Optimization

Optimization problems involves maximizing and minimizing real valued functions.

Example 1

A rectangular storage area is to be constructed along the side of a tall building. A security fence is required along the remaining 3 sides of the area. What is the maximum area that can be enclosed with 800m of fencing?

Solution



Area,

$$A = l \times w$$

$$= x(800 - 2x)$$

$$= 800x - 2x^2$$

$$\frac{dA}{dx} = 800 - 4x$$

For maximum/minimum area, $\frac{dA}{dx} = 0$

$$\Rightarrow 800 - 4x = 0$$

$$4x = 800$$

$$x = 200m$$

$$\frac{d^2A}{dx^2} = -4 < 0$$

$\Rightarrow x = 200m$ represent maximum area.

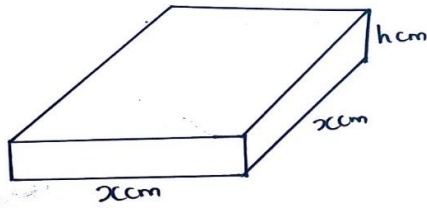
$$A_{\max} = 200(800 - 400)$$

$$= 80,000m^2$$

Example 2

A manufacturer wants to design an open box having a square base and a surface area of $108cm^2$. What dimensions will produce a box with maximum volume and hence find the maximum volume.

Solution



Surface area, $A = x^2 + 4xh = 108$

Volume, $V = x^2h$

$$\Rightarrow h = \frac{108 - x^2}{4x}$$

$$V = x^2 \left(\frac{108 - x^2}{4x} \right)$$

$$= 27x - \frac{x^3}{4}$$

$$\frac{dV}{dx} = 27 - \frac{3}{4}x^2$$

For maximum or minimum volume, $\frac{dV}{dx} = 0$

$$\Rightarrow 27 - \frac{3}{4}x^2 = 0$$

$$3x^2 = 108$$

$$x = \pm 6$$

$$\frac{d^2V}{dx^2} = -\frac{3}{2}x$$

When $x = -6$,

$$\frac{d^2V}{dx^2} = -\frac{3}{2}(-6)$$

$$= 9 > 0$$

$\Rightarrow x = -6$ represent minimum volume

When $x = 6$,

$$\frac{d^2V}{dx^2} = -\frac{3}{2}(6)$$

$$= -9 < 0$$

$\Rightarrow x = 6$ represent maximum volume

Therefore $x = 6$

$$h = \frac{108 - 36}{24} = 3\text{cm}$$

$$V_{\max} = 6 \times 6 \times 3$$

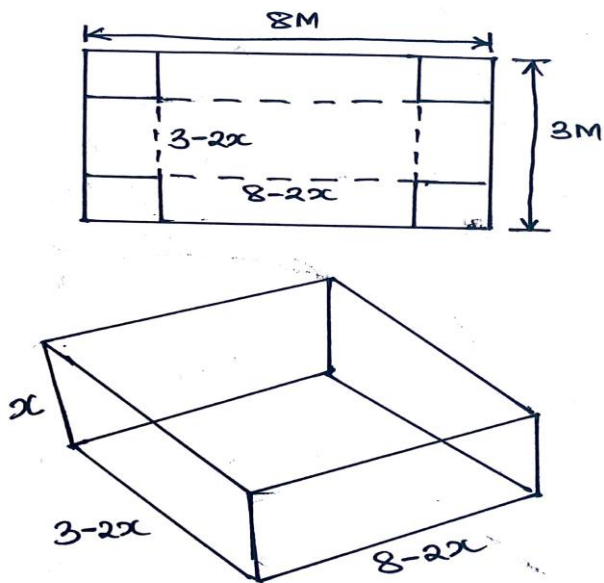
$$= 108\text{cm}^3$$

Example 3

An open topped box is to be constructed by removing an equal squares from each corner of a $3m$ by $8m$ rectangular sheet of aluminum and folding up sides. Find the volume of the largest such box.

Solution

Let xm be the length of a side of removed square



Volume,

$$V = x(3-2x)(8-2x)$$

$$= 4x^3 - 22x^2 + 24x$$

$$\frac{dV}{dx} = 12x^2 - 44x + 24$$

For maximum or minimum volume, $\frac{dV}{dx} = 12x^2 - 44x + 24 = 0$

$$\Rightarrow 4(x-3)(3x-2) = 0$$

$$\Rightarrow x = 3, \quad x = \frac{2}{3}$$

$$\frac{d^2V}{dx^2} = 24x - 44$$

When $x = 3$,

$$\frac{d^2V}{dx^2} = 72 - 44$$

$$= 28 > 0$$

$\Rightarrow x = 3$ represent minimum volume

When $x = \frac{2}{3}$,

$$\frac{d^2V}{dx^2} = 12 - 44$$

$$= -28 < 0$$

$\Rightarrow x = \frac{2}{3}$ represent maximum volume

$$V_{\max} = 4\left(\frac{2}{3}\right)^3 - 22\left(\frac{2}{3}\right)^2 + 24\left(\frac{2}{3}\right)$$

$$= 7.407m^3$$

Topic Summary

In this topic, you have learned how to determine the equation of tangent lines and normal lines to a curve, find maximum and minimum points to a curve, solve problems involving kinematics, related rates of change, optimization and finally sketching the curves for both the polynomials and asymptotes.

In summary, you learned that;

- In determining the equations of tangent and normal lines to curves, the two general equations are necessary

a) The equation of the tangent line at $P(x_0, y_0)$ to the curve $y = f(x)$ is given by

$$y - y_0 = m(x - x_0)$$

b) The equation of the normal line to the curve $y = f(x)$ at the point $P(x_0, y_0)$ is

given by $y - y_0 = -\frac{1}{m}(x - x_0)$

- In solving problems in kinematics where S - denotes displacement, V - velocity and a -

acceleration: $V = \frac{dS}{dt}$, $a = \frac{dV}{dt} = \frac{d^2S}{dt^2}$ and that

a) At zero velocity, ($V=0$), the object is said to be stationary and at this point, a maximum height is achieved.

b) At zero acceleration ($a=0$), the object is said to have a constant velocity and at this point, maximum velocity is achieved.

- A “related rates” problem is a problem in which one of the rates of change at a given instant is known and the other rate at that instant is to be determined. you find the equation that relates the two quantities and then use the appropriate technique of differentiation.

- Maximum and minimum points are known as extrema and occurs on a curve where $\frac{dy}{dx} = 0$. By checking the sign before and after the turning point, you will be able to classify it either as a maximum or a minimum point. A second derivative test can as well be used if suppose $(x_0, f(x_0))$ is a turning point of $f(x)$ and if:

a) $\frac{d^2y}{dx^2} < 0$, then $(x_0, f(x_0))$ is a maximum turning point.

b) $\frac{d^2y}{dx^2} > 0$, then $(x_0, f(x_0))$ is a minimum turning point.

- In sketching the polynomial curves, you need to find the general course of the curve by marking;

i. All the turning points (where $\frac{dy}{dx} = 0$).

ii. Intercepts ($x=0$ gives y -intercept and $y=0$ gives x -intercept).

- For asymptotic curves, the above procedure for polynomial curves applies in addition to finding the asymptotes.
- In optimization problems, the absolute minimum and or maximum of a function that depends on two variables given some constraint or relationship that the two variables must always satisfy are determined. Problems solved tend to center a round geometric objects such as boxes, square, rectangle, circle, cylinders etc.

